

Learnable Path-Complex Features and Unified Path–Cellular Message Passing for Link Prediction

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Motivation

1-WL bounds plain GNNs [?]. Topological lifts escape it by message-passing over a richer object:

- **MPSN** [?]: biased to *cliques*.
- **CWN** [?]: biased to *induced cycles*.
- **PCN** [?]: simple paths, no substructure bias.

PCN leaves two problems on the table:

1. Symmetric scatter-sum init forces $A \rightarrow B \rightarrow C$ and $C \rightarrow B \rightarrow A$ to receive *identical* embeddings.
2. Lift is monolithic: paths *or* rings. Link prediction needs the closed-cycle signal that paths alone discard.

Our Contributions

1. **Position-aware learnable init.** $k+1$ position-specific linear maps per k -cell: strictly subsumes the scatter-sum and breaks orientation symmetry at $t=0$.
2. **Unified path and cellular complex.** Path- and ring-cells coexist at dim 2; canonicalised edges route messages between them for free.
3. **Theory (U-PWL).** U-PWL $\geq$ PWL *and* U-PWL $\geq$ CWL(k -IC); strictly $>$ 1-WL.
4. **Cora ablation, 10 seeds.** Each modification cuts seed variance; unified lift achieves the best mean test AUC.

Big Takeaway

Unified path- and ring-cells with a learnable, order-aware init give PCN strictly greater expressivity and the best Cora AUC — at zero kernel-level cost.

- **Greater expressivity:** proved via U-PWL theory.
- **Lower variance:** -31% test-AUC std across 10 seeds.
- **Zero kernel cost:** same MP architecture as PCN.

Background: Path Complexes

A *simple k -path* is an ordered tuple $\sigma = (v_0, \dots, v_k)$ of distinct vertices with $\{v_i, v_{i+1}\} \in \mathcal{E}$; write $P_k(G)$ for the full set. The boundary alternates end-deletion, $\partial e_{i_0 \dots i_p} = \sum_q (-1)^q e_{i_0 \dots \hat{i}_q \dots i_p}$ [?], and PCN message-passes over the faces $\mathcal{B}(\sigma)$ together with the upper adjacencies $\mathcal{N}_\uparrow(\sigma)$.

$\mathcal{S}_2 \subset \mathcal{A}_2$: path complexes strictly enrich the 2-cell vocabulary.

The Deterministic-Init Problem

PCN initialises cell features by a symmetric aggregation, $h_\sigma^{(0)} = \bigoplus_{v \in \sigma} x_v$ with $\bigoplus \in \{\text{sum, mean}\}$ [?], which immediately collapses orientation: $h_{(A,B,C)}^{(0)} = h_{(C,B,A)}^{(0)}$. The damage is permanent, since later permutation-invariant aggregation also discards the signed boundary, so the source/target asymmetry that link decoders rely on is erased before message passing even starts.

Notation

$P_k(G), R_L(G)$	k -paths; induced cycles
$\sigma = (v_0, \dots, v_k)$	ordered k -cell
$\mathcal{B}, \mathcal{N}_\uparrow$	faces; upper-adjacents
$W_i^{(k)}$	pos.-aware init weight
$\mathcal{C}(G)$	unified path and ring complex

Method 1: Position-Aware Learnable Init

Replace the scatter-sum with $k+1$ position-specific maps:

$$h_\sigma^{(0)} = \sum_{i=0}^k W_i^{(k)} x_{v_i}$$

Each node contributes once per cell, so the asymptotic cost is unchanged. Sharing $W_i^{(k)} = W$ recovers a linear projection of the scatter-sum, so the new init strictly subsumes the deterministic one; for unequal weights, paths $\sigma \neq \sigma'$ with the same multi-set of endpoints receive *distinct* features, something no symmetric $\bigoplus$ can produce. The first vs. last endpoint encodes source vs. target, exactly the asymmetry link decoders already exploit. The price is $(k+1) d d_m$ parameters per k -cell ($d=64$ here); the init still runs in $\mathcal{O}(|P_k| d)$.

Method 2: Unified Path and Cellular Complex

Stack ring-cells *on top of* the path complex at dim 2:

$$\mathcal{C}_2(G) = \underbrace{P_2(G)}_{2\text{-paths}} \cup \underbrace{R_L(G)}_{\text{induced cycles, } L \leq 7}$$

Path- and ring-cells have different boundary rules but share a single edge representation: a ring and a path on the same edge resolve to the *same* canonical edge cell, so upper-adjacency mixes ring $\leftrightarrow$ path messages for free.

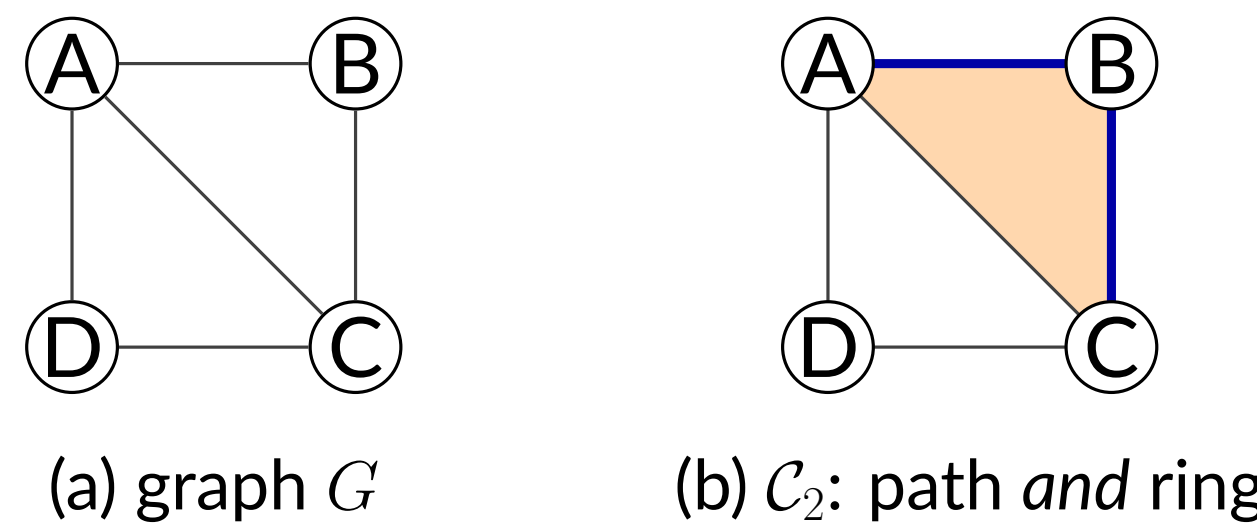


Figure 1. **Unified path and cellular complex.** On triangle ABC , the **ring-cell** (orange fill) and the **path-cell** $A \rightarrow B \rightarrow C$ (blue arrows) coexist and share edges AB, BC via canonicalised IDs, so $\mathcal{N}_\uparrow$ routes messages between them at zero kernel cost.

Why this targets link prediction

Common neighbours and short cycles drive link prediction (Adamic–Adar, SEAL). Path-cells expose only the *open* neighbourhoods of (u, v) , whereas ring-cells expose the *closed* ones, supplying exactly the signal that path-only PCN misses.

Boundary rules differ; adjacency unifies

The path boundary is the signed alternating end-deletion $\partial(v_0, \dots, v_k) = \sum_q (-1)^q (v_0, \dots, \hat{v}_q, \dots, v_k)$, while the ring boundary collects the cycle's edges unsigned (the standard cellular ∂_b). Both reduce to the *same* canonical edge IDs, so a single $\mathcal{N}_\uparrow$ graph carries both kinds of message with no extra MP kernel. On Cora this adds +26% dim-2 cells; each triangle contributes 1 ring and 3 paths, so rings supply the triadic-closure signal that paths alone cannot encode.

Construction Recipe

1. Enumerate $P_2(G)$ and $R_{\leq 7}(G)$ separately (one-off preprocessing).
2. Canonicalise every edge by sorting endpoint pairs; this is the *single* representation any 1-cell receives.
3. Build boundary and upper-adjacency tensors over the merged dim-2 set $\mathcal{C}_2(G) = P_2 \cup R_{\leq 7}$.
4. Run standard PCN message passing; the position-aware init

Worked Example: Symmetry Breaking

Take graph G : $a-b-c$ with one-hot features $x_a=e_1$, $x_b=e_2$, $x_c=e_3$. Compare reversed paths $\sigma=(a,b,c)$ and $\sigma'=(c,b,a)$.

Deterministic init collapses orientation:

$$h_\sigma^{(0)} = e_1 + e_2 + e_3 = h_{\sigma'}^{(0)}.$$

Learnable init with $W_0^{(2)}=I$, $W_1^{(2)}=\mathbf{0}$, $W_2^{(2)}=2I$:

$$h_\sigma^{(0)} = I e_1 + \mathbf{0} e_2 + 2I e_3 = e_1 + 2 e_3, \\ h_{\sigma'}^{(0)} = I e_3 + \mathbf{0} e_2 + 2I e_1 = e_3 + 2 e_1 \neq h_\sigma^{(0)}.$$

Orientation therefore **survives** initialization and flows through every later message-passing step.

Encoder → Decoder Pipeline

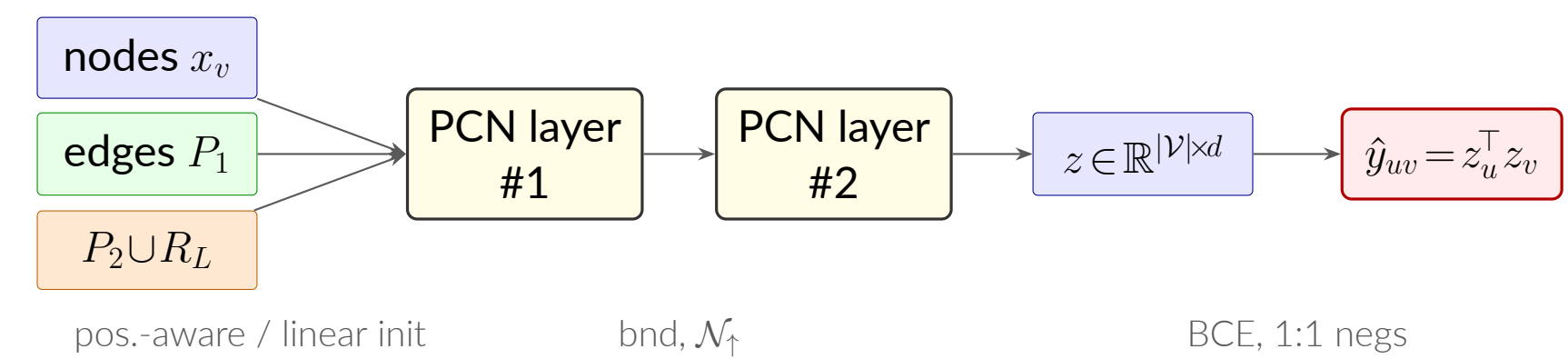


Figure 2. Cell features (dim-0,1,2) are initialised (linear for nodes, *position-aware* for paths and rings) and refined by two PCN message-passing layers; the node-level output feeds a dot-product link decoder trained with BCE.

Message Passing on the Unified Complex

Each layer aggregates boundaries and upper-adjacent cells:

$$m_B(\sigma) = \sum_{\tau \in \mathcal{B}(\sigma)} M_B(h_\tau^{(t)}), \quad m_\uparrow(\sigma) = \sum_{\tau \in \mathcal{N}_\uparrow(\sigma)} M_\uparrow(h_\tau^{(t)}), \\ h_\sigma^{(t+1)} = \text{UP}(h_\sigma^{(t)}, m_B(\sigma), m_\uparrow(\sigma)).$$

$M_B, M_\uparrow, \text{UP}$: GIN-style MLPs [?].

Drop-in. The same sums silently mix **ring** and **path** messages with the *kernel unchanged*.

Theoretical Results

U-PWL: WL test on $\mathcal{C}(G)$

$$c_\sigma^{(t+1)} = \text{Hash}(c_\sigma^{(t)}, \{\{c_\tau^{(t)}\}\}_{\mathcal{B}(\sigma)}, \{\{c_\tau^{(t)}\}\}_{\mathcal{N}_\uparrow(\sigma)})$$

- **Thm 1.** U-PWL $\geq$ PWL.
- **Thm 2.** U-PWL $\geq$ CWL(k -IC) at dim 2.
- **Cor.** U-PWL strictly $>$ 1-WL; not less than 3-WL.

Why: adding ring-cells preserves every PCN path-cell and adjacency, so U-PWL refines both PWL and CWL(k -IC) [? ? ?].

Cost & Complexity on Cora

Both modifications stay **cheap**: learnable init adds parameters only; the unified lift pays a one-off preprocessing cost. Per-epoch latency stays in the path-only budget.

Config	Lift (s)	Epoch (ms)	Params
Path, Det.	22.0	28.5	428,544
Path, Learn.	22.0	37.5	703,680
Unified, Det.	62.8	33.0	428,544
Unified, Learn.	62.8	43.2	703,680

- **Learnable init:** +64% params, +32% epoch.
- **Unified lift:** +26% dim-2 cells; 2.9× preprocessing, *runs once*.
- **Asymptotic:** $\mathcal{O}(n \cdot b^{\max(k,L)})$ [?].

Cora Link Prediction: Setup & Results

- **Data:** Cora [?], $|\mathcal{V}|=2708$, $|\mathcal{E}|=5278$.
- **Split:** 85/5/10 [?], 1:1 negatives.
- **Model:** 2-layer encoder, hidden 64, dropout 0.3, dot decoder; $k_{\max}=2$, $L=7$.
- **Train:** Adam lr 0.01, 200 epochs, early-stop on val AUC. Mean $\pm$ std over 10 seeds.

Complex	Init	Val AUC	Test AUC
Path	Det.	0.7344 $\pm$ 0.0274	0.7113 $\pm$ 0.0239
Path	Learn.	0.7354 $\pm$ 0.0230	0.7123 $\pm$ 0.0164
Unified	Det.	0.7366 $\pm$ 0.0270	0.7132 $\pm$ 0.0165
Unified	Learn.	0.7335 $\pm$ 0.0213	0.7111 $\pm$ 0.0175

Table 1. ROC-AUC, 10 seeds. **Learnable alone:** test std 0.0239 $\rightarrow$ 0.0164 (-31%). **Unified alone:** best mean test AUC. **Stacked:** partially redundant on Cora.

Per-Seed Variance

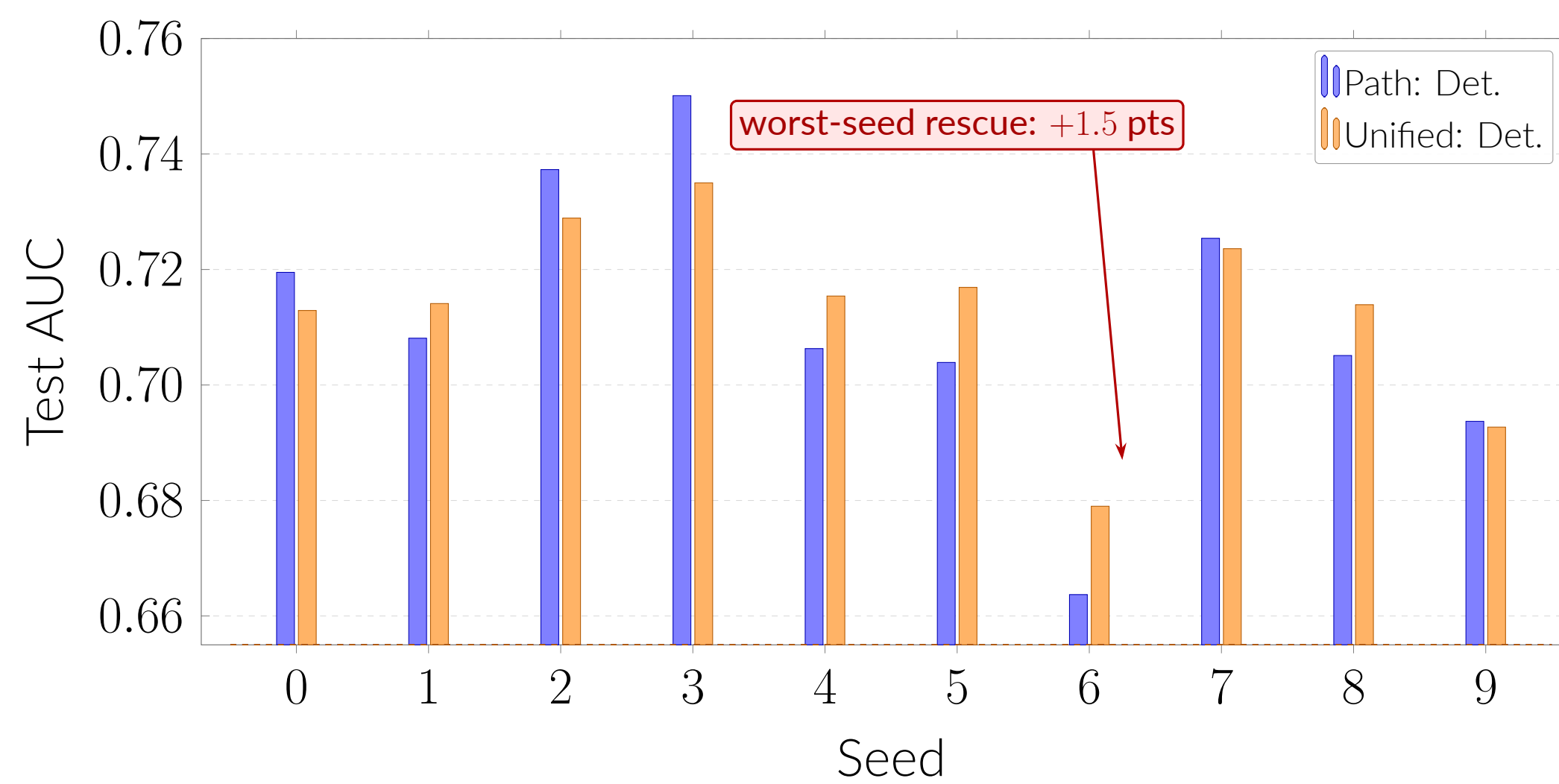


Figure 3. Per-seed test AUC; dashed lines mark the 10-seed means (Path 0.7113, Unified 0.7132). The unified lift lifts the worst-case seeds the most (seed 6: 0.664 $\rightarrow$ 0.679), consistent with the stronger triadic signal supplied by ring-cells.

Limitations & Future Work

Limitations. Cora gains are modest ($<1\%$ AUC); stacked modifications interact; closing the gap to SEAL / NBFNet requires larger benchmarks.

Future.

- Clique-cells alongside paths and rings.
- Larger benchmarks: OGBL-COLLAB / OGBL-PPA.
- Learnable gating to rebalance triangle over-weighting (each 3-cycle is one ring and three paths).

References